ENGR 232 — All Homework Problems & Solutions

HW1 — Ch.2: Atomic Structure & Bonding

2.2 Cr has 4 isotopes: 4.34% of ^{50}Cr (49.9460 amu), 83.79% of ^{52}Cr (51.9405 amu), 9.50% of ^{53}Cr (52.9407 amu), 2.37% of ^{54}Cr (53.9389 amu). Confirm avg atomic weight = 51.9963 amu.

Sol: $\bar{A} = \sum f_i A_i$:

$$\bar{A} = (0.0434)(49.9460) + (0.8379)(51.9405)$$

$$+(0.0950)(52.9407) + (0.0237)(53.9389)$$

$\Rightarrow$ **= 51.9963 amu** ✓

2.5 (a) Grams in 1 amu? (b) Atoms in 1 lb-mol?

Sol: (a) $\dfrac{1 \text{ g/mol}}{6.022 \times 10^{23} \text{ atoms/mol}} \Rightarrow$ **1.66×10^{-24} g/amu**

(b) 453.6 g/lb$_m$ $\times$ 6.022×10^{23} atoms/mol $\Rightarrow$ **2.73×10^{26} atoms/lb-mol**

2.15 Force of attraction between K^+ and O^{2-} ions separated by 1.5 nm.

Sol: $F_A = \dfrac{A}{r^2}$, where $A = \dfrac{Z_1 Z_2 e^2}{4\pi\varepsilon_0}$, $Z_1 = 1$, $Z_2 = 2$, $r = 1.5$ nm $\Rightarrow$ **$F_A = 2.05 \times 10^{-10}$ N**

2.22 (a) Main differences between ionic, covalent, metallic bonding? (b) Pauli exclusion principle?

Sol: (a) *Ionic*: electrostatic attraction between oppositely charged ions. *Covalent*: electron sharing so each atom achieves stable config. *Metallic*: ion cores shielded/glued by sea of valence electrons. (b) Each electron state holds ≤ 2 electrons with opposite spins.

2.25 % ionic character of TiO_2, ZnTe, CsCl, InSb, $MgCl_2$.

Sol: $\%IC = \left[1 - e^{-0.25(X_A - X_B)^2}\right] \times 100$

TiO_2: $X_{Ti} = 1.5$, $X_O = 3.5 \Rightarrow$ **$\approx 63.2\%$**
ZnTe: $X_{Zn} = 1.6$, $X_{Te} = 2.1 \Rightarrow$ **$\approx 6.1\%$**
CsCl: $X_{Cs} = 0.7$, $X_{Cl} = 3.0 \Rightarrow$ **$\approx 74.3\%$**
InSb: $X_{In} = 1.7$, $X_{Sb} = 1.9 \Rightarrow$ **$\approx 1.0\%$**
$MgCl_2$: $X_{Mg} = 1.2$, $X_{Cl} = 3.0 \Rightarrow$ **$\approx 55.5\%$**

HW2 — Ch.3: Crystal Structures

3.3 Show for BCC that $a = \dfrac{4R}{\sqrt{3}}$.

Sol: Using body diagonal of BCC unit cell: the diagonal $= 4R$ and $= a\sqrt{3}$. Therefore: $\Rightarrow a = \dfrac{4R}{\sqrt{3}}$ ✓

3.7 Density of iron (BCC, $R = 0.124$ nm, $A = 55.85$ g/mol).

Sol: $\rho = \dfrac{nA}{V_C N_A}$, BCC: $n = 2$, $V_C = a^3 = \left(\dfrac{4R}{\sqrt{3}}\right)^3$

$$a = \frac{4(0.124)}{\sqrt{3}} = 0.2866 \text{ nm}$$

$\Rightarrow$ **$\rho = 7.90$ g/cm^3** (exp: 7.87 g/cm^3)

3.24 Point indices for all atoms in BCC unit cell.

Sol: Corner atoms at: 000, 100, 110, 010, 001, 101, 111, 011. Body-center atom: $\Rightarrow$ **$\frac{1}{2}\frac{1}{2}\frac{1}{2}$**

3.43 Indices for two crystallographic planes (one parallel to x, z, intercepts y at $b/2$; one intercepts at $a/2$, $-b/2$, c).

Sol: Plane 1: $A = \infty a$, $B = b/2$, $C = \infty c$ $\Rightarrow h = 0, k = 2, l = 0 \Rightarrow$ **(020)**

Plane 2: $A = a/2$, $B = -b/2$, $C = c \Rightarrow h = 2, k = -2, l = 1 \Rightarrow$ **$(2\bar{2}1)$**

3.63 Interplanar spacing d_{111} for Mo (BCC, $R = 0.1363$ nm).

Sol: $a = \dfrac{4R}{\sqrt{3}} = \dfrac{4(0.1363)}{\sqrt{3}} = 0.3148$ nm

$d_{hkl} = \dfrac{a}{\sqrt{h^2 + k^2 + l^2}}$, so $d_{111} = \dfrac{0.3148}{\sqrt{3}} \Rightarrow$ **= 0.1817 nm**

HW3 — Ch.4: Imperfections in Solids

4.2 Hypothetical metal: $N_v = 2.8 \times 10^{24}$ m^{-3} at 750°C, $\rho = 5.60$ g/cm^3, $A = 65.6$ g/mol. Find vacancy fraction.

Sol: $N = \dfrac{N_A \rho}{A} = \dfrac{(6.022 \times 10^{23})(5.60 \times 10^6)}{65.6} = 5.14 \times 10^{28}$ m^{-3}

$\Rightarrow N_v/N = \dfrac{2.8 \times 10^{24}}{5.14 \times 10^{28}} = 5.44 \times 10^{-5}$

4.4 Vacancies/m^3 in Fe at 850°C ($Q_v = 1.08$ eV, $\rho = 7.65$ g/cm^3, $A = 55.85$ g/mol).

Sol: $N_v = N \exp\left(-\dfrac{Q_v}{kT}\right)$, $T = 1123$ K, $N = 8.24 \times 10^{28}$ m^{-3} $\Rightarrow$ **$N_v = 1.18 \times 10^{24}$ m^{-3}**

4.14 Composition (wt%) of 6 at% Pb – 94 at% Sn alloy.

Sol: $C'_{Pb} = \dfrac{C_{Pb} A_{Pb}}{C_{Pb} A_{Pb} + C_{Sn} A_{Sn}} \times 100$

$A_{Pb} = 207.2$, $A_{Sn} = 118.71 \Rightarrow$ **$C_{Pb} = 10.0$ wt%, $C_{Sn} = 90.0$ wt%**

4.16 Composition (at%) of alloy with 98 g Sn + 65 g Pb.

Sol: $n'_{Sn} = 98/118.71 = 0.825$ mol, $n'_{Pb} = 65/207.2 = 0.314$ mol $\Rightarrow$ **$C_{Sn} = 72.4$ at%, $C_{Pb} = 27.6$ at%**

4.26 Au atoms/cm^3 in 10 wt% Au – 90 wt% Ag alloy ($\rho_{Au} = 19.32$, $\rho_{Ag} = 10.49$ g/cm^3).

Sol: Using Equation 4.21: $C''_{Au} = \dfrac{C_{Au} N_A}{\dfrac{C_{Au} A_{Au}}{\rho_{Au}} + \dfrac{C_{Ag} A_{Ag}}{\rho_{Ag}}} \Rightarrow$ **$= 3.36 \times 10^{21}$ atoms/cm^3**

HW4 — Ch.5: Diffusion

5.8 kg of H_2 passing/hr through Pd sheet: thickness 5 mm, area 0.20 m^2, 500°C, $D = 1.0 \times 10^{-8}$ m^2/s, $C_A = 2.4$ kg/m^3, $C_B = 0.6$ kg/m^3.

Sol: $J = -D\dfrac{\Delta C}{\Delta x}$, $M = JAt$

$J = (1.0 \times 10^{-8})\dfrac{2.4 - 0.6}{5 \times 10^{-3}} = 3.6 \times 10^{-6}$ kg/m^2/s

$M = J \times A \times 3600$ s $\Rightarrow$ **$= 2.6 \times 10^{-3}$ kg/h**

5.15 N concentration at 1 mm depth after 10 h in Fe at 700°C ($C_s = 0.1$ wt%, $D = 2.5 \times 10^{-11}$ m^2/s).

Sol: $C_x = C_s\left[1 - \text{erf}\left(\dfrac{x}{2\sqrt{Dt}}\right)\right]$

$\dfrac{x}{2\sqrt{Dt}} = \dfrac{10^{-3}}{2\sqrt{(2.5 \times 10^{-11})(36000)}} = 0.527$

erf(0.527) $\approx 0.5436 \Rightarrow$ **$C_x = 0.046$ wt% N**

5.19 D for Zn in Cu at 650°C ($D_0 = 2.4 \times 10^{-5}$ m^2/s, $Q_d = 189{,}000$ J/mol).

Sol: $D = D_0 \exp\left(-\dfrac{Q_d}{RT}\right)$, $T = 923$ K $\Rightarrow$ **$D = 4.8 \times 10^{-16}$ m^2/s**

5.24 Ag in Cu: $D_{650} = 5.5 \times 10^{-16}$, $D_{900} = 1.3 \times 10^{-13}$ m^2/s. (a) Find D_0, Q_d. (b) D at 875°C.

Sol: (a) Using two equations $\ln D = \ln D_0 - Q_d/RT$: $\Rightarrow$ **$Q_d = 196{,}700$ J/mol**, $\Rightarrow$ **$D_0 = 7.5 \times 10^{-5}$ m^2/s**

(b) $T = 1148$ K: $\Rightarrow$ **$D = 8.3 \times 10^{-14}$ m^2/s**

HW5 — Ch.6: Mechanical Properties of Metals

6.3 Elastic strain in Al specimen (10×12.7 mm cross section) under 35,500 N.

Sol: $\sigma = F/A_0 = 35500/(127 \times 10^{-6}) = 279.5$ MPa

$\varepsilon = \sigma/E = 279.5 \times 10^6/(69 \times 10^9) \Rightarrow$ **$= 4.05 \times 10^{-3}$**

6.6 Elongation of Ti wire ($d = 3.0$ mm, $L_0 = 2.5 \times 10^4$ mm) under 500 N.

Sol: $\Delta l = \dfrac{FL_0}{A_0 E}$, $A_0 = \pi(1.5 \times 10^{-3})^2 = 7.07 \times 10^{-6}$ m^2, $E_{Ti} = 107$ GPa

$\Rightarrow$ **$\Delta l = 16.6$ mm**

6.10 Elongation of steel alloy specimen ($d = 10$ mm, $L_0 = 75$ mm) under 20,000 N.

Sol: $\sigma = F/A_0 = 20000/(\pi(5 \times 10^{-3})^2) = 255$ MPa

From stress-strain curve, $\varepsilon \approx 0.0012$ (elastic region)

$\Delta l = \varepsilon L_0 = 0.0012 \times 75 \Rightarrow$ **= 0.090 mm**

6.28 Steel bar ($L_0 = 300$ mm, sq. cross-section 4.5 mm side). (a) Load for $\Delta l = 0.45$ mm. (b) Deformation after release.

Sol: (a) $\varepsilon = \Delta l/L_0 = 0.45/300 = 0.0015$. From Fig. 6.22: $\sigma \approx 300$ MPa (elastic).

$F = \sigma A_0 = 300 \times 10^6 \times (4.5 \times 10^{-3})^2 \Rightarrow$ **= 6075 N**

(b) $\Rightarrow$ **No permanent deformation** (elastic only)

HW6 — Ch.7 & 8: Dislocations, Fracture, Fatigue

7.14 Critical resolved shear stress for Ag: tensile along [001], slip on (111) in $[\bar{1}01]$, initiated at $\sigma = 1.1$ MPa.

Sol: λ = angle([001],$[\bar{1}01]$): $\cos\lambda = \dfrac{(0)(-1) + (0)(0) + (1)(1)}{\sqrt{1}\sqrt{2}} = \dfrac{1}{\sqrt{2}}$

ϕ = angle([001],[111]): $\cos\phi = \dfrac{1}{\sqrt{3}}$

$\tau_{crss} = \sigma \cos\phi \cos\lambda = 1.1 \cdot \dfrac{1}{\sqrt{3}} \cdot \dfrac{1}{\sqrt{2}} \Rightarrow$ **= 0.45 MPa**

7.20 Why are small-angle grain boundaries less effective at blocking slip than high-angle boundaries?

Sol: $\Rightarrow$ **Less crystallographic misalignment at small-angle boundaries $\Rightarrow$ less change in slip direction $\Rightarrow$ less interference.**

7.30 Pre-cold-worked radius of Cu wire ($r_d = 10$ mm, ductility 25%EL).

Sol: From Fig. 7.19c: 25%EL $\Rightarrow\approx$ 11%CW.

$\%CW = \left(\dfrac{r_0^2 - r_d^2}{r_0^2}\right) \times 100 \Rightarrow 0.11 = 1 - (10/r_0)^2$

$\Rightarrow$ **$r_0 = 10/\sqrt{0.89} = 10.6$ mm**

8.4 Max allowable surface crack length in polystyrene ($\sigma = 1.25$ MPa, $\gamma_s = 0.50$ J/m^2, $E = 3.0$ GPa).

Sol: $a = \dfrac{2E\gamma_s}{\pi\sigma^2} = \dfrac{2(3.0 \times 10^9)(0.50)}{\pi(1.25 \times 10^6)^2} \Rightarrow$ **$= 6.1 \times 10^{-4}$ m = 0.61 mm**

8.22 Brass fatigue data. (a) S-N plot. (b) Fatigue strength at 5×10^5 cycles. (c) Fatigue life at 200 MPa.

σ_a (MPa)	Cycles
310	2×10^5
223	1×10^6
168	1×10^7
143	1×10^8
127	1×10^9

(b) $\Rightarrow$ **≈ 250 MPa at 5×10^5 cycles**

(c) $\Rightarrow$ **$\approx 2 \times 10^6$ cycles at 200 MPa**

HW7 — Ch.9: Phase Diagrams I

9.11 Is it possible to have a Cu-Ni alloy in equilibrium with liquid (20 wt% Ni) and α (37 wt% Ni)?

Sol: $\Rightarrow$ **No.** A single tie line in the $\alpha + L$ region cannot intersect the phase boundaries at both 20 wt% Ni (liquidus ≈1200°C) and 37 wt% Ni (solidus ≈1230°C).

9.14 50 wt% Pb–50 wt% Mg alloy cooled from 700°C to 400°C. (a) T of first solid? (b) Composition of first solid? (c) T of complete solidification? (d) Composition of last liquid?

Sol: (a) $\Rightarrow$ **≈ 560°C** (intersection of vertical line with liquidus)

(b) $\Rightarrow$ **21 wt% Pb–79 wt% Mg** (tie line at 560°C, solidus intercept)
(c) $\Rightarrow \approx 465$**°C** (eutectic isotherm)
(d) $\Rightarrow \approx 67$ **wt% Pb–33 wt% Mg** (eutectic composition)

HW8 — Ch.9: Phase Diagrams II

9.19 90 wt% Ag–10 wt% Cu in $\beta + L$ region, liquid at 85 wt% Ag. (a) T? (b) β composition? (c) Mass fractions?

Sol: (a) Tie line at 85 wt% Ag on liquidus: $\Rightarrow \approx 850$**°C**
(b) Solidus intersection: $\Rightarrow \approx 95$ **wt% Ag**
(c) Lever rule, $C_0 = 90$, $C_L = 85$, $C_\beta = 95$ wt% Ag: $W_\beta = (90-85)/(95-85) \Rightarrow = 0.5$, $W_L = 0.5$

9.22 A–B alloy: $C_0 = 55$ wt% B, $W_\alpha = W_\beta = 0.5$, $C_\beta = 90$ wt% B. Find C_α.

Sol: $W_\alpha = (C_\beta - C_0)/(C_\beta - C_\alpha) = 0.5$
$(90-55)/(90-C_\alpha) = 0.5 \Rightarrow 90 - C_\alpha = 70 \Rightarrow$ $C_\alpha = 20$ **wt% B**

9.53 Carbon concentration for total ferrite fraction = 0.94.

Sol: $W_\alpha = (0.76 - C_0)/(0.76 - 0.022)$, solve: $0.94 \times 0.738 = 0.76 - C_0$
$\Rightarrow C_0 \approx 0.067$ **wt% C**

9.58 Mass fractions of proeutectoid ferrite and pearlite in 0.25 wt% C alloy.

Sol: $W_p = \dfrac{C_0' - 0.022}{0.74} = \dfrac{0.25 - 0.022}{0.74} \Rightarrow = 0.308$

$W_{\alpha'} = \dfrac{0.76 - C_0'}{0.74} = \dfrac{0.76 - 0.25}{0.74} \Rightarrow = 0.689$

9.61 Alloy with $W_{\alpha'} = 0.20$, $W_p = 0.80$: find carbon concentration.

Sol: From $W_p = (C_0 - 0.022)/0.74 = 0.80$:
$C_0 - 0.022 = 0.592 \Rightarrow C_0 = 0.61$ **wt% C**

HW9 — Ch.10 & 12: Phase Trans. & Ceramics

10.3 Critical radius for homogeneous nucleation in Cu at 849°C ($\Delta H_f = -1.77 \times 10^9$ J/m^3, $\gamma = 0.200$ J/m^2).

Sol: $r^* = -\dfrac{2\gamma}{\Delta G_v} = \dfrac{2\gamma T_m}{|\Delta H_f|(T_m - T)}$

$= \dfrac{2(0.200)(1358)}{1.77 \times 10^9 (1358 - 1122)} \Rightarrow = 1.29 \times 10^{-9}$ **m** $= 1.29$ **nm**

10.8 Avrami kinetics: $n = 2.5$, $y = 0.40$ at $t = 200$ min. Find recrystallization rate.

Sol: $y = 1 - e^{-kt^n}$, solve for k:

$k = -\dfrac{\ln(1-0.40)}{200^{2.5}} = \dfrac{0.5108}{1.789 \times 10^6} = 2.855 \times 10^{-7}$

$t_{0.5} = \left(\dfrac{\ln 2}{k}\right)^{1/n} = \left(\dfrac{0.693}{2.855 \times 10^{-7}}\right)^{0.4} \Rightarrow \approx$ 214 **min**

Rate $= 1/t_{0.5} \Rightarrow \approx 4.7 \times 10^{-3}$ **min**$^{-1}$

12.16 Density of FeO (rock salt structure, $r_{Fe^{2+}} = 0.077$ nm, $r_{O^{2-}} = 0.140$ nm).

Sol: $n' = 4$, $a = 2(r_{Fe^{2+}} + r_{O^{2-}}) = 2(0.217) = 0.434$ nm

$\rho = \dfrac{n'(\sum A)}{V_C N_A} = \dfrac{4(55.85 + 16.00)}{(4.34 \times 10^{-8})^3 (6.022 \times 10^{23})} \Rightarrow$ $= 5.84$ **g/cm**^{3}

12.18 CdS: $a = 0.582$ nm, $\rho = 4.82$ g/cm^3. How many Cd^{2+} and S^{2-} per unit cell?

Sol: $n' = \dfrac{\rho V_C N_A}{\sum A} = \dfrac{4.82(0.582 \times 10^{-7})^3 (6.022 \times 10^{23})}{112.41 + 32.06}$

$\Rightarrow n' \approx 4$ **Cd^{2+} and 4 S^{2-} per unit cell**

12.29 Schottky defect fraction in NaCl at 801°C ($Q_s = 2.3$ eV).

Sol: $\dfrac{N_s}{N} = \exp\left(-\dfrac{Q_s}{2kT}\right)$, $T = 1074$ K, $k = 8.62 \times 10^{-5}$ eV/K

$= \exp\left(-\dfrac{2.3}{2(8.62 \times 10^{-5})(1074)}\right) \Rightarrow = 4.03 \times 10^{-6}$

HW10 — Ch.14 & 18: Polymers & Electrical

14.5 Polypropylene MW data: compute (a) $\bar{M}_n$, (b) $\bar{M}_w$, (c) degree of polymerization.

Range	M_i	x_i	$x_i M_i$	$w_i M_i$
8k–16k	12k	0.05	600	240
16k–24k	20k	0.16	3200	2000
24k–32k	28k	0.24	6720	5600
32k–40k	36k	0.28	10080	10800
40k–48k	44k	0.20	8800	11880
48k–56k	52k	0.07	3640	5720

(a) $\bar{M}_n = \sum x_i M_i \Rightarrow = 33{,}040$ **g/mol**
(b) $\bar{M}_w = \sum w_i M_i \Rightarrow = 36{,}240$ **g/mol**
(c) Polypropylene repeat unit: $m = 3(12.01) + 6(1.008) = 42.08$ g/mol
$DP = \bar{M}_n/m = 33040/42.08 \Rightarrow \approx 785$

18.3 Max length of 4-mm-dia Al wire for $R \leq 2.5\ \Omega$ ($\sigma_{Al} = 3.8 \times 10^7$ (Ω-m)$^{-1}$).

Sol: $R = \dfrac{l}{\sigma A}$, $A = \pi(2 \times 10^{-3})^2 = 1.257 \times 10^{-5}$ m^2
$l = R\sigma A = 2.5 \times 3.8 \times 10^7 \times 1.257 \times 10^{-5} \Rightarrow = 1194$ **m**

18.5 Cu wire: $d = 3$ mm, $L = 2$ m, $\sigma_{Cu} = 6.0 \times 10^7$ (Ω-m)$^{-1}$.

Sol: $A = \pi(1.5 \times 10^{-3})^2 = 7.07 \times 10^{-6}$ m^2
(a) $R = L/(\sigma A) = 2/(6 \times 10^7 \times 7.07 \times 10^{-6}) \Rightarrow$ $= 4.7 \times 10^{-3}$ **Ω**
(b) $I = V/R = 0.05/4.7 \times 10^{-3} \Rightarrow = 10.6$ **A**
(c) $J = I/A = 10.6/7.07 \times 10^{-6} \Rightarrow = 1.5 \times 10^6$ **A/m^2**
(d) $\mathcal{E} = V/L = 0.05/2 \Rightarrow = 0.025$ **V/m**

18.10 Drift velocity of electrons in Ge at 1000 V/m ($\mu_e = 0.38$ m^2/V-s). Time to traverse 25 mm?

Sol: (a) $v_d = \mu_e \mathcal{E} = 0.38 \times 1000 \Rightarrow = 380$ **m/s**
(b) $t = l/v_d = 0.025/380 \Rightarrow = 6.6 \times 10^{-5}$ **s**

18.21 Intrinsic carrier concentration of PbTe at room temp ($\sigma = 500$ (Ω-m)$^{-1}$, $\mu_e = 0.16$, $\mu_h = 0.075$ m^2/V-s).

Sol: $\sigma = n_i e(\mu_e + \mu_h)$

$n_i = \dfrac{\sigma}{e(\mu_e + \mu_h)} = \dfrac{500}{(1.6 \times 10^{-19})(0.16 + 0.075)}$
$\Rightarrow = 1.33 \times 10^{22}$ **m^{-3}**